DATA ITEM DESCRIPTION

Title: RELIABILITY CENTERED MAINTENANCE (RCM) LOGIC TREE ANALYSIS (LTA) WITH SUPPORTING RATIONALE AND JUSTIFICATION REPORT

Number: DI-SESS-80984A
AMSC Number: N9167
DTIC Applicable: N/A
Office of Primary Responsibility: SH/SEA 04RM
Applicable Forms: N/A

Approval Date: 20100923
Limitation: N/A
GIDEP Applicable: N/A

Use/Relationship:
The Reliability Centered Maintenance (RCM) Logic Tree Analysis (LTA) with supporting rationale and justification defines the need for a periodic maintenance (PM) task and provides descriptions and periodicities.

This Data Item Description (DID) contains the format, content, preparation instructions and intended use for the data deliverable resulting from the work task described in 5.1.5 of MIL-STD-3034.

This DID is related to DI-SESS-80979A, RCM Master System and Subsystem Index (MSSI); DI-SESS-80994A, RCM Functional Block Diagram (FBD); DI-SESS-80981A, RCM Functional Failure Analysis(FFA)Report; DI-SESS-80983A, RCM Additional Functionally Significant Item (AFSI) Selection Report; DI-SESS-80982A, RCM Functionally Significant Items (FSI) Index; DI-SESS-80980A, RCM Failure Modes and Effects Analysis (FMEA) Report; DI-SESS-80985A, RCM Servicing and Lubrication Analysis (SLA) Report; DI-SESS-81829, RCM Corrective Maintenance (CM) Development Report; DI-SESS-80989A, RCM Inactive Equipment Maintenance (IEM) Requirement Analysis Report; DI-SESS-80986A, RCM Maintenance Requirements Index (MRI); DI-SESS-80988A, RCM Task Definition Report; DI-SESS-80987A, RCM Procedure Validation Report.

This DID supersedes DI-MNTY-80984.

Requirements:

1. Reference documents. The applicable issue of documents cited herein, including approval dates and dates of any applicable amendments, notices, and revisions, shall be as cited in the contract.

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2. Format. This report shall be in Contractor's format and shall be presented in the electronic database specified in the contract.

3. Content. This report shall contain all of the information specified in the sample form of Figure 1 of this DID as follows:

3.1 Block 1. ESWBS number. Duplicate each Expanded Ship Work Breakdown Structure (ESWBS) entry on the Functionally Significant Item (FSI) index, block 8. Each Item is to have its own analysis.

3.2 Block 2. Nomenclature. Duplicate the entry on the FSI index, block 9.

3.3 Block 3. Ship class. Duplicate the entry on the FSI index, block 3.

3.4 Block 4. Prepared by. Enter the analyst's name and the date.

3.5 Block 5. Reviewed by. Enter the first level reviewer's name and the date.

3.6 Block 6. Approved by. Reserved for the maintenance coordinating activity approval signature and the date.

3.7 Block 7. Revision. Enter Original, A, B, or C, sequentially and the date.

3.8 Block 8. Functional failure and failure mode(s). Enter functional failures and related failure modes receiving a YES in block 12 of the Failure Modes and Effects Analysis (FMEA); number each as numbered in the FMEA.

3.9 Block 9. Criticality analysis. Enter "Y", "N" or "NA" to signify a yes, no or not applicable answer to each of the first three logic tree questions. Possible alternatives are:

Questions			Criticality Class
1	2	3	Assignment
Y	Y	N/A	A
Y	N	Y	B
Y	N	N	C
N	N/A	N/A	D

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3.10. Block 10. Criticality class. Enter A, B, C, or D, based on the answers in block 9 above. These letters identify the four main branches of the logic tree as follows:

a. Class A – Operating safety

b. Class B – Operating capability/mission

c. Class C – Other regular functions (economics driven)

d. Class D – Hidden or infrequent functions

3.11 Block 11. Periodic Maintenance (PM) task. Enter "Y" or "N" to signify a yes or no answer to questions 4, 5, 6, or 7 in the logic tree. If a task identified during this phase is a Servicing or Lubrication task, the analysis for this functional failure and failure mode combination should be transferred to Phase 6.

a. If the criticality class for this failure is "A", the Condition Directed or Time Directed task identified must be applicable, and must reduce the probability of occurrence of the functional failure to an acceptable level; cost is not a consideration. If no task is available, a safety related design change recommendation must be identified in block 13.

b. If the criticality class for this failure is "B", the Condition Directed or Time Directed task identified must be applicable, and must reduce the risk (product of probability and severity) of the occurrence of the functional failure to an acceptable level; cost is not a consideration.

c. If the criticality class for this failure is "C", the Condition Directed or Time Directed task identified must be applicable, and must be cost effective.

d. If the criticality class for this failure is "D", the Condition Directed or Time Directed task identified must be applicable, and must meet the rule of effectiveness associated with the consequence of the functional failure. If no PM suitable task is available, move to question 8 of the logic tree.

3.12 Block 12. Failure Finding (FF) Task. Enter "Y" or "N" signifying a yes or no answer to question 8 in the logic tree.

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3.13 Block 13. Redesign recommendation. Enter "Y" if redesign was identified for reducing the risk of this failure; otherwise enter "N"

3.14 Block 14. Task description. Enter a brief description of the applicable and effective task(s).

3.15 Block 15. Periodicity. Enter the periodicity for the accomplishment of tasks in block 14.

3.16 Block 16. Serial numbers. Enter four-segment serial number as follows:

Segment 1 - Enter the developing organization abbreviation followed by a slant (/).

b. Segment 2 - For developers, enter the development authorization number followed by a slant (/); for other development activities, assign a development number followed by a slant (/).

c. Segment 3 - Enter the number 120 to indicate the logic tree analysis form, followed by a slant (/).

d. Segment 4 - Enter the ESWBS number from block 1.

3.17 Decision Logic tree questions rationale and justification documentation. The rationale and justification for the answers to the decision logic tree questions shall be identified in free-form, simple narrative on plain paper sheets attached to the Logic Tree Analysis. The rationale and justification for the questions pertinent to each functional failure analyzed should be identified by functional failure and associated failure mode number and description.

4. End of DI-SESS-80984A.

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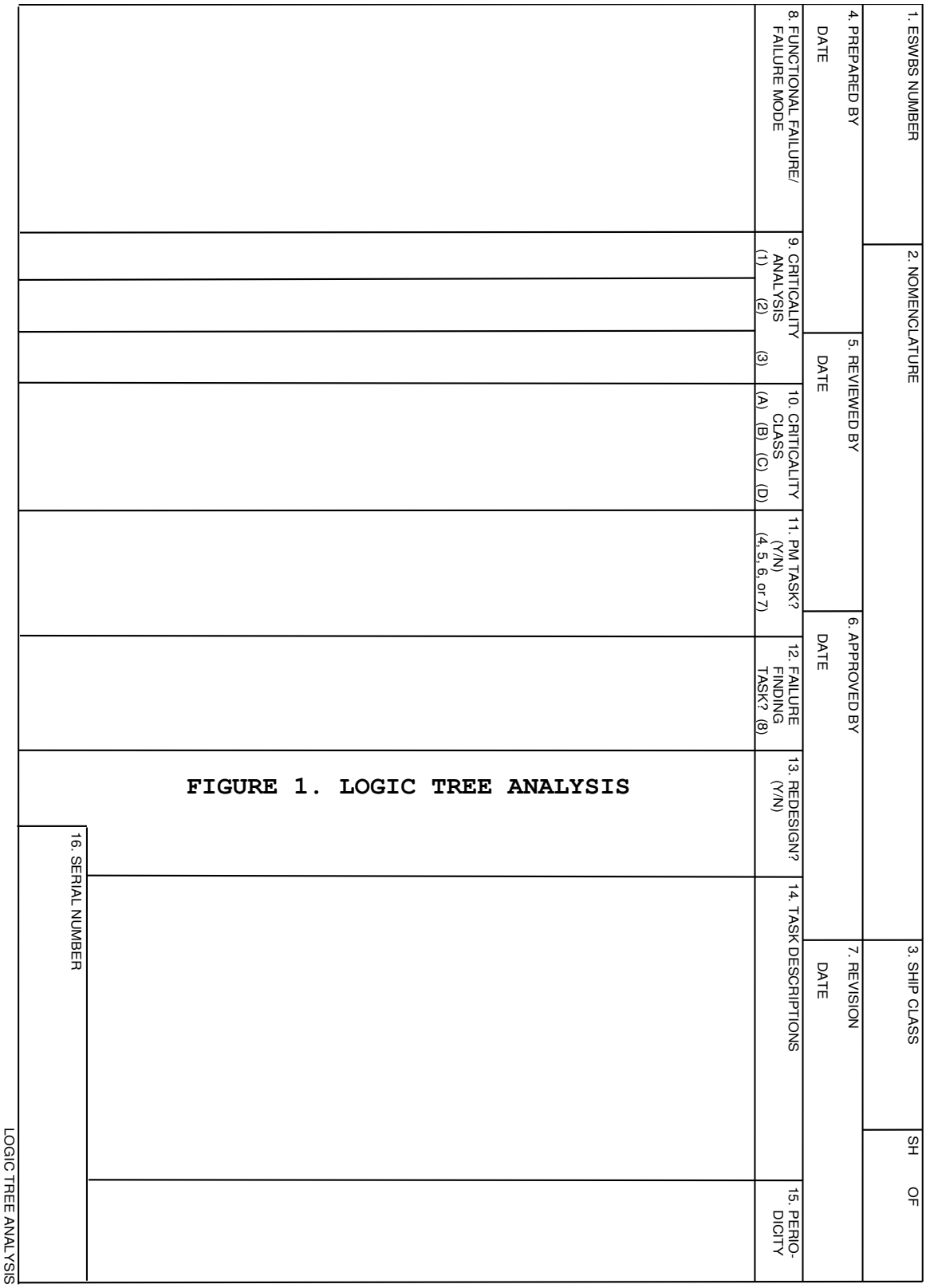


FIGURE 1. LOGIC TREE ANALYSIS

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